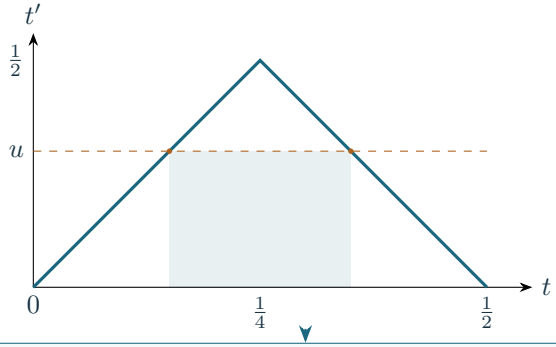
A source energy becomes a triangle entropy

Exact coefficient Cauchy inequality

The positive tail expansion gives, for every real spectral parameter,

$$\frac{|q_k(\lambda)|^2}{q_k(\lambda_0)} \quad \leq \quad q_k(\lambda^2/\lambda_0).$$

The folded angle has a uniform distribution



Shown: $\delta = \frac{1}{4}$.

$t' = \frac{1}{\pi} \arctan |R_\delta(t)|$.

For every δ,
$|\{t : t' > u\}| = \frac{1}{2} - u$.

Proved uniformly for every one-sided coefficient

$$\frac{1}{N} \log \frac{\sup_{\sigma \in [-2,2]} |a_N(q;\sigma)|^2}{p_N(q)} \quad \leq \quad C(q_x, q_y, \tfrac{1}{2} - q_x - q_y) + o(1).$$

Both level parities; original unused rows and protected zeros retained.

Remaining global comparison

Keep all original face loads and combine estimates on each shared flow.

$$\max_{q \in \mathcal{P}_D^{\mathrm{NC}}} \sum_e \min\left\{ h_{\mathrm{env},e}(q_e), \frac{h_e(q_e) + g_e(q_e)}{2} \right\} \quad < \quad \Phi_D.$$

This sufficient inequality is explicit; universal validity is open.